Adam Shebani

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EDUCATION

University of Wisconsin-Madison, College of Engineering

Degree: Bachelor of Science, Computer Engineering - GPA: 3.9/4.0 *Expected:* December 2027

ENGINEERING EXPERIENCE

Silicon Badgers – RTL Team Designer

- Profiled Qwen3.5-2B inference in llama.cpp by compute-graph operation, attributing ~90% of runtime to matrix multiplication and defining the accelerator's target workload.
- Designed a pipelined, parameterizable systolic array in SystemVerilog with INT4/INT8 MAC elements for the attention and MLP layers, verified against a NumPy golden model.

Digital System Design & Synthesis (SystemVerilog) – ECE 551 UW-Madison

- Synthesized SystemVerilog control logic in Synopsys Design Compiler, applying timing constraints and closing timing at 400 MHz.
- Integrated a maze-solving robot SoC with UART, SPI inertial sensing, PID motor control, and FSM navigation in synthesizable SystemVerilog, using a wall-following strategy to solve mazes.

Embedded Systems and Microcontrollers (C) – ECE 353 UW-Madison

- Developed embedded C firmware on an ARM Cortex-M (PSoC 6) using interrupt-driven design with timers, GPIO, PWM, and ADC peripherals.
- Built concurrent I/O functionality using FreeRTOS to coordinate sampling, communication, and display updates without polling.

PERSONAL PROJECTS

Adam16 – 16-bit RISC CPU (SystemVerilog, FPGA/ASIC)

- Designed the RTL for a 16-bit CPU with a custom ISA, FSM-based control unit, and datapath; verified with a 33-case directed testbench achieving full functional coverage.
- Ran on an Artix-7 FPGA (Basys 3) via Vivado, and synthesized to a 32 nm standard-cell library in Design Compiler at 333 MHz (3 ns), passing gate-level simulation in Questa.

SFP+ Carrier Board (Altium, 4-layer PCB)

- Designed a 4-layer high-speed PCB with a controlled-impedance stackup: 100 Ω differential and 50 Ω single-ended grounded coplanar waveguide routed to SMA connectors for signal readout.
- Implemented via stitching and a separate chassis ground for signal integrity and EMI, and generated Gerber, BOM, and NC drill outputs for fabrication.

Wearable Motion Tracker (Infineon PSoC 6, C, FreeRTOS)

- Brought up six peripherals over SPI, I2C, and PDM under FreeRTOS on an ARM Cortex-M4: IMU, magnetometer, pressure, 60 GHz radar, MEMS mic, and QSPI flash.
- Logged and streamed multi-sensor data to train a CNN gesture classifier, deployed on-device at 84% accuracy with 200ms inference latency.

SKILLS

- **Languages:** C, C++, Python, Java, SystemVerilog, TCL
- **Design & Verification:** RTL design, STA, gate-level simulation, ModelSim, Questa
- **EDA/FPGA:** Synopsys Design Compiler, Vivado, Quartus
- **Embedded:** ARM Cortex-M, FreeRTOS, UART, SPI, I2C, PWM
- **PCB & Lab:** Altium, KiCad, SPICE, oscilloscope, soldering
- **Tools:** Git, GitHub, Linux